THE ASSOCIATED CHAIN COMPLEX FUNCTOR S IS EXACT

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-s-exact (source lines 3864-3867); the proof environment begins on the second line after it and is its own proof, closing at source line 3871.

Editorial note. The author's proof is the single sentence 'Clear from Lemma lemma-abelian'; the card reproduces it as written, carries the cited lemma as context, and records in the pairing note that no further detail is supplied by the author.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: the associated chain complex functor s is exact.

Allowed standard prerequisites (editorial).

- The associated chain complex functor s. The functor s sending a simplicial object U to the chain complex with s(U)_n = U_n and differential the alternating sum of the face maps.. It is the functor of the statement (source line 3866).
- Exactness of the simplicial construction. The author's lemma-abelian, from which the exactness of s is read off (retained as context).. It is the whole justification (source lines 3869-3870).

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category.*

(1) *The categories Simp($\mathcal{A}$) and CoSimp($\mathcal{A}$) are abelian.*
(2) *A morphism of (co)simplicial objects $f : A \to B$ is injective if and only if each $f_n : A_n \to B_n$ is injective.*
(3) *A morphism of (co)simplicial objects $f : A \to B$ is surjective if and only if each $f_n : A_n \to B_n$ is surjective.*
(4) *A sequence of (co)simplicial objects*

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact at B if and only if each sequence

$$A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i$$

is exact at B_i.

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Proof. Pre-additivity is easy. A final object is given by $U_n = 0$ in all degrees. Existence of direct products we saw in Lemmas 0.2 and 0.3. Kernels and cokernels are obtained by taking termwise kernels and cokernels. □

Lemma 0.2. *If U and V are simplicial objects in the category $\mathcal{C}$, and if $U \times V$ exists, then we have*

$$\mathrm{Mor}(W, U \times V) = \mathrm{Mor}(W, U) \times \mathrm{Mor}(W, V)$$

for any third simplicial object W of $\mathcal{C}$.

Proof. Omitted. □

Lemma 0.3. *If U and V are cosimplicial objects in the category $\mathcal{C}$, and if $U \times V$ exists, then we have*

$$\mathrm{Mor}(W, U \times V) = \mathrm{Mor}(W, U) \times \mathrm{Mor}(W, V)$$

for any third cosimplicial object W of $\mathcal{C}$.

Proof. Omitted. □

Lemma 0.4. *The functor s is exact.*

Proof. Clear from Lemma 0.1. □